7 Rings

Definition 7.1.
A (commutative) ring $(R, +, \times)$ consists of a nonempty set R endowed with two operations: addition $+$, and multiplication $\times$, satisfying the following axioms.

(A1) $(R, +)$ is an abelian group. We denote the neutral element of $+$ by 0_R or simply 0, and the inverse of an element a for addition by $-a$. Thus

$$a + (-a) = 0.$$

(A2) $a \times b = b \times a \quad \forall a, b \in R$ (multiplication is commutative).

(A3) $(a \times b) \times c = a \times (b \times c) \quad \forall a, b, c \in R$ (associativity of multiplication).

(A4) $a \times (b + c) = a \times b + a \times c \quad \forall a, b, c \in R$ (distributivity of $\times$ with respect to $+$).

We say that R has an identity element (called a ring with identity) if $\exists$ an element $1_R \in R$ (or simply denoted 1) such that

$$a \times 1_R = a, \ \forall a \in R.$$

In the notations we often omit the sign $\times$ and write $ab = a \times b$. Also we denote

$$a - b = a + (-b).$$

Example 7.1.

1. $(\mathbb{Z}, +, \times)$ is a ring with identity 1.
2. $(\mathbb{Q}, +, \times)$, $(\mathbb{R}, +, \times)$, and $(\mathbb{C}, +, \times)$ are rings with identity 1.
3. For $m \geq 1$, $(\mathbb{Z}/m\mathbb{Z}, +, \times)$ is a ring with identity the class $\bar{1}$ of 1.
4. $(\mathbb{Q}[X], +, \times)$ is a ring with identity the constant polynomial $f(X) = 1$.

Lemma 7.1.
Let R be a ring. Then the followings hold.

1. $0_R a = a 0_R = 0_R, \ \forall a \in R.$
2. $(-a)b = a(-b) = -(ab), \ \forall a, b \in R.$
3. $(-a)(-b) = ab, \ \forall a, b \in R.$
4. *If R has an identity element 1_R then* $-a = (-1_R)a, \ \forall a \in R.$

Proof. See Problem Sheet 4 exercise 1. □

Definition 7.2.
Let R be a ring. A nonzero element $a \in R \setminus \{0_R\}$ is called a *zero divisor* if $\exists b \in R \setminus \{0_R\}$ such that

$$ab = 0_R.$$

The notation $R \setminus \{0_R\}$ denotes the set of nonzero elements of R.

Example 7.2.

1. $\bar{2} \in \mathbb{Z}/4\mathbb{Z}$ is a zero divisor since $\bar{2} \times \bar{2} = \bar{4} = \bar{0}$.
2. $\mathbb{Z}$ has no zero divisors since
$$\{ab = 0, a, b \in \mathbb{Z}\} \implies \{a = 0 \quad \text{or} \quad b = 0\}.$$
3. $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors if and only if n is a prime integer.

   Proof. Indeed if n is not a prime: $n = n_1 n_2$ with $1 < n_i < n$, hence $\bar{n} = 0 = \bar{n}_1 \bar{n}_2$ and $0 \neq \bar{n}_i$ is a zero divisor, $1 \leq i \leq 2$. Conversely assume $n = p$ is a prime integer and $\bar{a}\bar{b} = \bar{0} = \bar{p} = \overline{ab}$, then p divides ab and hence either p divides a (which is equivalent to $\bar{a} = \bar{0}$) or p divides b (which is equivalent to $\bar{b} = \bar{0}$). □
4. $\mathbb{Q}[X]$ has no zero divisors.

   Proof. We prove that the product of two nonzero polynomials in $\mathbb{Q}[X]$ is a nonzero polynomial. Suppose $f(X) = a_0 + a_1 X + \ldots + a_n X^n \neq 0$, thus $\deg(f) = n \geq 0$ and $a_n \neq 0$; and $g(X) = b_0 + b_1 X + \ldots + b_m X^m \neq 0$, thus $\deg(g) = m \geq 0$ and $b_m \neq 0$. Then
   $$f(X)g(X) = a_0 b_0 + (a_0 b_1 + a_1 b_0)X + \ldots + a_n b_m X^{n+m} \neq 0,$$
   since $a_n b_m \neq 0$. □

From now on when we assume that a ring R has an identity element 1_R, we suppose $1_R \neq 0_R$ [if $0_R = 1_R$ then all elements of R equal 0_R: $a = (1_R)a = (0_R)a = 0_R$].

Definition 7.3.
Let R be a ring with an identity element 1. An element $u \in R$ is called a *unit* if $\exists v \in R$ such that
$$uv = 1.$$
We denote v by u^{-1} and call it the *inverse* of u for multiplication.

Lemma 7.2.
Let R be a ring with identity 1. The set of units of R is denoted
$$R^\times = \{u \in R : u \text{ unit}\}.$$
The set $R^\times$ is closed under multiplication, and $(R^\times, \times)$ is an abelian group.

Proof. First $1 \in R^\times$ since $1 \times 1 = 1$, hence 1 is a unit. Let $u_1, u_2 \in R^\times$, there exists $v_1, v_2 \in R$ such that
$$u_1 v_1 = u_2 v_2 = 1.$$
Then
$$1 = (u_1 v_1)(u_2 v_2) = (u_1 u_2)(v_1 v_2)$$
hence $u_1 u_2$ is a unit and
$$(u_1 u_2)^{-1} = v_1 v_2 = u_1^{-1} u_2^{-1}.$$
Further let $u \in R^\times$, there exists $v \in R$ such that $uv = 1$ hence $v \in R^\times$ is a unit and $v = u^{-1}$ is the inverse of u. Note that the group law $\times$ of $R^\times$ is associative by the axiom (A3) of a ring. □

Example 7.3.

1. Let $u \in \mathbb{Z}$ be a unit, then $\exists v \in \mathbb{Z}$ such that $uv = 1$ which implies $u = 1$ or $u = -1$. Hence $\mathbb{Z}^\times = \{1, -1\}$.
2. $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \ : \ \gcd(a, n) = 1\}$.

   Proof. Indeed $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times \iff \exists \bar{b} \in \mathbb{Z}/n\mathbb{Z} : \bar{a}\bar{b} = \overline{ab} = \bar{1} \iff \exists k \in \mathbb{Z} : ab = nk + 1 \iff \exists k \in \mathbb{Z} : ab - nk = 1 \iff \gcd(a, n) = 1$. □
3. $(\mathbb{Q}[X])^\times = \mathbb{Q}^\times = (\mathbb{Q} \setminus \{0\}, \times)$ (see Lemma 6.3).

Definition 7.4.
A ring is called an *integral domain* if it has no zero divisors.

Example 7.4.

1. $\mathbb{Z}$ is an integral domain
2. $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is a prime integer.

Definition 7.5.
A ring F with identity is called a *field* if $F^\times = F \setminus \{0\}$, in other words F is a field if every nonzero element of F is a unit.

Example 7.5.

1. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are fields.
2. $\mathbb{Z}$ and $\mathbb{Q}[X]$ are not fields: 2 has no inverse in $\mathbb{Z}$ and X has no inverse in $\mathbb{Q}[X]$.
3. $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if $n = p$ is a prime integer.

   Proof. Indeed if $\bar{0} \neq \bar{k} \in \mathbb{Z}/n\mathbb{Z}$ then $\bar{k} \in (\mathbb{Z}/n\mathbb{Z})^\times \iff \gcd(k, n) = 1$ (see example 2 in Examples 7.3). Hence every non zero element of $\mathbb{Z}/n\mathbb{Z}$ is a unit if and only if n has no positive divisor k with $1 < k < n$, which is precisely the definition of n being a prime. □

Definition 7.6.
Let p be a prime integer. The field $\mathbb{Z}/p\mathbb{Z}$ is denoted $\mathbb{F}_p$ and called the *finite field with p element*.

Lemma 7.3.
A field F is necessarily an integral domain.

Proof. Let $a, b \in F$ with $ab = 0$. We show either $a = 0$ or $b = 0$. Suppose for example $a \neq 0$, we show that $b = 0$. Since $a \neq 0$, a is a unit of F and $\exists a^{-1} \in F$ such that $a^{-1}a = 1$. Thus $a^{-1}(ab) = (a^{-1}a)b = 1b = b = 0$. □

Definition 7.7.
A subset S of a ring R is called a *subring* if $(S, +)$ is a subgroup of $(R, +)$ and S is closed under multiplication, i.e. $a, b \in S$ implies $ab \in S$.

Example 7.6.

1. $\mathbb{Q}$ is a subring of $\mathbb{R}$, $\mathbb{R}$ is a subring of $\mathbb{C}$, $\mathbb{Q}$ is a subring of $\mathbb{Q}[X]$.
2. For $m \in \mathbb{Z}$ set $m\mathbb{Z} = \{ma : a \in \mathbb{Z}\}$. Then $m\mathbb{Z}$ is a subring of $\mathbb{Z}$ (without identity if $|m| > 1$).
3. Let $D \in \mathbb{Z}$ be a non zero square free integer, i.e. D is not divisible by a square. Equivalently, $D = p_1 p_2 \dots p_t$, $D = -1$, or $D = -p_1 p_2 \dots p_t$, where $p_1, p_2, \dots, p_t$ are distinct primes.

Set
$$\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} : a, b \in \mathbb{Q}\} \subset \mathbb{C}.$$
Then $\mathbb{Q}(\sqrt{D})$ is a subring of $\mathbb{C}$. Indeed, $0 = 0 + 0\sqrt{D} \in \mathbb{Q}(\sqrt{D})$,
$$(a_1 + b_1\sqrt{D}) + (a_2 + b_2\sqrt{D}) = (a_1 + a_2) + (b_1 + b_2)\sqrt{D} \in \mathbb{Q}(\sqrt{D}),$$
$$-(a_1 + b_1\sqrt{D}) = -a_1 + (-b_1)\sqrt{D} \in \mathbb{Q}(\sqrt{D})$$
and
$$(a_1 + b_1\sqrt{D})(a_2 + b_2\sqrt{D}) = (a_1a_2 + b_1b_2D) + (a_2b_1 + a_1b_2)\sqrt{D} \in \mathbb{Q}(\sqrt{D})$$
if $a_1 + b_1\sqrt{D}, a_2 + b_2\sqrt{D} \in \mathbb{Q}(\sqrt{D})$.

- In fact $\mathbb{Q}(\sqrt{D})$ is a field. Indeed let $0 \neq a + b\sqrt{D} \in \mathbb{Q}(\sqrt{D})$ which implies $a \neq 0$ or $b \neq 0$: if $a = b = 0$ then $a + b\sqrt{D} = 0 + 0\sqrt{D} = 0$. We have
$$(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \neq 0$$
[if $a^2 - Db^2 = 0$; which is equivalent to $a^2 = Db^2$, then both a and b are non zero (if one of them is zero then the other is also zero which contradicts $a \neq 0$ or $b \neq 0$), and $D = \frac{a^2}{b^2} = (\frac{a}{b})^2$ would be a square], and
$$(a + b\sqrt{D})\frac{a - b\sqrt{D}}{a^2 - Db^2} = (a + b\sqrt{D})(\frac{a}{a^2 - Db^2} - \frac{b}{a^2 - Db^2}\sqrt{D}) = 1$$
hence $a + b\sqrt{D}$ is a unit since $(\frac{a}{a^2-Db^2} - \frac{b}{a^2-Db^2}\sqrt{D}) \in \mathbb{Q}(\sqrt{D})$.

4. With the notations in (3) above set
$$\mathbb{Z}[\sqrt{D}] = \{a + b\sqrt{D} : a, b \in \mathbb{Z}\} \subset \mathbb{Q}(\sqrt{D}).$$
Then $\mathbb{Z}[\sqrt{D}]$ is a subring of $\mathbb{Q}(\sqrt{D})$ (check this fact).
$$\mathbb{Z}[\sqrt{-1}] = \mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$$
is called the *ring of Gaussian integers*.
5. Define
$$\mathbb{Z}[X] = \{a_0 + a_1X + \ldots + a_nX^n : a_i \in \mathbb{Z}, n \geq 0\} \subset \mathbb{Q}[X].$$
Then $\mathbb{Z}[X]$ is a subring of $\mathbb{Q}[X]$ (check this fact).

Example 7.7.

1. Show that $(\mathbb{Z}[\sqrt{-1}])^\times = \{1, -1, i, -i\}$. [Hint: Show and use the fact that if $a + ib \in \mathbb{Z}[i]$ is a unit then $|a + ib| = a^2 + b^2 = 1$.]
Show that the group $((\mathbb{Z}[\sqrt{-1}])^\times, \times)$ is isomorphic to $\mathbb{Z}/4\mathbb{Z}$.
2. $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$.
The group of units $(\mathbb{Z}[\sqrt{2}])^\times$ is infinite. Indeed $1 + \sqrt{2}$ is a unit since
$$(\sqrt{2} - 1)(1 + \sqrt{2}) = 1.$$

Further $(1+\sqrt{2})^n$ is a unit $\forall n \geq 1$ since

$$(\sqrt{2}-1)^n(1+\sqrt{2})^n = [(\sqrt{2}-1)(1+\sqrt{2})]^n = 1^n = 1.$$

The set

$$\{(1+\sqrt{2})^n : n \geq 1\}$$

is infinite as

$$(1+\sqrt{2})^{n_1} \neq (1+\sqrt{2})^{n_2} \quad \text{if} \quad n_1 > n_2.$$

Indeed otherwise $(1+\sqrt{2})^{n_1-n_2} = 1$ which is not possible since $1+\sqrt{2} > 1$ and $n_1 - n_2 > 0$.